

Modeling Customer Lifetime Value in Buyer-Seller Relationships

Chang Xue

International Economics & Trade
Shandong University of Finance
Jinan, P. R. China
e-mail: changxue20284@163.com

Abstract—Estimating customer lifetime value is becoming increasingly important in order to identify and invest on prospective profitable customers. This paper reviews three research streams in this field: models for calculation of CLV; models of customer base analysis; normative models of CLV in order to evaluate the usefulness of customer lifetime value as a metric for customer selection and marketing resource allocation. Lifetime value can be used to inform the nature and quality of relationship that an organization might have with customers. Customers who are selected on the basis of their lifetime value provide higher profits in future periods than do customers selected on the basis of several other customer-based metrics. Managers can improve profits by designing marketing communications that maximize CLV.

Keywords—customer lifetime value; customer equity; customer retention; customer relationship

I. INTRODUCTION

Customer lifetime value (CLV) is being recognized as one of the most important measures of the worth of a customer, as it takes into account not only the customer's value but the expected value over their projected lifetime with your company. The assessment of customer lifetime values in buyer-seller relationships has a long tradition in marketing theory and practice.

Customer lifetime value for a firm is the net profit or loss to the firm from a customer over the entire life of transactions of that customer with the firm. Hence the lifetime value of a customer for a firm is the net of the revenues obtained from that customer over the lifetime of transactions with that customer minus the cost of attracting, selling, and servicing that customer, taking into account the time value of money [1].

II. MODELS OF CLV

Many different ways of measuring and calculating CLV have been developed, depending upon different trading situations and purposes. In the literature, CLV also appears under other names such as customer equity and customer profitability. Many models have been proposed in CLV literature dealing with all kinds of issues related to CLV. The following selection of models provides summaries of some key models addressing some major research opportunities in CLV research and applications. Based on the threefold stream of research related to CLV, this section is divided into three corresponding categories: models for calculation of

CLV; models of customer base analysis; normative models of CLV.

A. Models for calculation of CLV

This category includes models that are specifically formulated to calculate the CLV and/or extend this calculation to obtain optimal methods of resource allocation to optimize CLV. These are applied models and more relevant to practitioners who wish to use CLV as a basis for making strategic or tactical decisions.

1) Basic structural model of CLV

$$CLV = \sum_{i=1}^n \frac{(R_i - C_i)}{(1 + d)^{i-0.5}}$$

Where, i = the period of cash flow from customer transaction; R_i =revenue from the customer in period i ; C_i =total cost of generating the revenue R_i in period i ; n = the total number of periods of projected life of the customer under consideration.

In this model, it is assumed that all cash flows take place at the end of a time period.

This model identifies a class of different CLV models based on the net present value (NPV) of the future cash flows from customers. This basic idea of NPV captures the essence of such models. Some important features of these models are that they assume a particular time of cash flow that is the same in each time period, they apply only to customers who are doing business with the firm, they ignore past as well as prospective customers, they ignore acquisition costs, they do not consider a number of important factors such as the stochastic nature of the purchase process and timing of cash flows, and they are very simple and therefore easy to use.

2) Customer migration model

Dwyer described a customer migration model for CLV analysis [2]. He proposed that customers can be broadly divided into two groups: always-a-share and lost-for-good. In the former, customers may rely on several vendors and can adjust their share of business done with each whereas in the latter category, customers have made long-term commitments to a vendor because switching vendors is costly and assets dedicated to the transaction cannot be redeployed easily.

The model proposed by Dwyer has an advantage over the basic structural model in that it considers the probabilistic nature of customer purchases. Based on past purchase behavior, the purchase probabilities are updated. Hence a customer might still be considered retained by a firm even if

she does not purchase in any particular period. This feature in Dwyer's model makes it more realistic. The model is still simple enough to be used in practice.

Despite its strengths, Dwyer's model does have some important weaknesses. The time period is fixed and the sale and cash flow are assumed to occur in the same period and at the same time in each period. Although the model assumptions make good sense for certain types of businesses such as magazine subscriptions, other types of businesses where the revenue stream is more uncertain might not be able to justify the use of this model. In addition, the propensities of purchase in each period are assumed to be the same and depend only on the period of last purchase.

3) *Optimal resource allocation models*

Blattberg and Deighton propose a managerially relevant model for finding the optimal balance between spending on customer acquisition and customer retention in order to maximize CLV [3]. They further propose that customer equity gains and losses should be tracked against marketing programs. Their model consists of two parts:

a) *Optimal level of acquisition spending*

$$a = (\text{ceiling rate})[1 - \exp(-k_1 \$A)]$$

The net contribution margin from acquiring a prospect in the first year = $\alpha \$m - \A

Where $\$A$ =the acquisition expenditure per prospect; a =the acquisition rate obtained as a result of acquisition expenditure; ceiling rate=the limit to attraction of new customers; k_1 =parameter controlling the shape of the exponential curve.

b) *Optimal level of retention spending*

The following equation describes how customer equity depends on retention spending:

$$r = (\text{ceiling rate})[1 - \exp(-k_2 \$R)]$$

Year y contribution from retention = $\alpha [\$m - \$R/r]$

Where $\$R$ =the retention expenditure per prospect; r = the retention rate obtained as a result of the retention expenditure; k_2 =the parameter controlling the shape of the exponential curve; $\$m$ =the margin we earned in the year we acquired the customer; it is assumed to remain the same in each year thereafter.

The annual values for each year of the customer's projected life are summed up, add the value of a first year customer, discount to the present value at a rate of return appropriate for marketing investments, and so obtain the amount of customer equity attributable to that customer. The optimal level of retention and acquisition spending is the level at which customer equity is maximized.

This model uses CLV as a basis for making optimal decisions of marketing resource allocation, unlike other models discussed previously where the objective had been to calculate CLV only. Further, this model takes into account the expenses on customer acquisition for calculating CLV. Although this model adds new dimensions to the existing CLV models, it continues to suffer from some of the same old weaknesses that other models have had. In particular, the time period for cash flow remains constant and cash flows are assumed to occur at the same time in each period. In

addition, the model does not consider customer acquisition and retention jointly to maximize customer equity.

Berger and NasrBechwati recently proposed a general approach to address the problem of optimal allocation of the promotion budget to maximize customer equity [4]. Their method uses decision calculus to maximize customer equity by considering optimal promotional expenditure allocation decisions between customer acquisition, customer retention, and other promotional options. In addition, they offer a series of applications of their approach under different market conditions.

4) *Customer relationship models*

Pfeifer and Carraway proposed that a general class of mathematical models called Markov Chain Models (MCM) is appropriate for modeling customer relationships [5]. The authors believed that MCM models are very flexible. MCM can be used to model both customer retention and customer migration situations. Customer migration refers to such a situation where a customer might remain inactive for some periods and still be treated as a retained customer on his/her return. The probabilistic nature of MCM allows accounting for the inherent stochasticity in customer relationships.

B. *Models of customer base analysis*

Such models take into account the past purchase behavior of the entire customer base in order to come up with probabilities of purchase in the next time period. These models take into consideration the stochastic behavior of customers in making purchases and therefore these models look at each customer individually in order to compute the probability of purchase in the next time period. Models in this category can provide input for the calculation of CLV.

1) *Pareto/NBD model*

Schmittlein, Morrison, and Colombo proposed a model, called the Pareto/NBD model, which calculates the probability that a customer is still active [6]. The model requires the number and timing of customers' previous transactions as input. Using this model, firms can identify and count the customers who are still active.

The Pareto/NBD model is applicable in contexts where the time when the customer becomes inactive is unknown to the analyst and the customer can make any number of purchases, at any time, and can become inactive any time. This model can be very useful to firms having few long-term customers. This model provides a sophisticated way to get these probabilities of a customer being active in each time period. The probabilities thus obtained can then be used to calculate CLV.

In the calculation of CLV, the most critical part is the determination of the number of customers who are still active and the number of customers who will be active in each future time period. This model provides a sophisticated way to get these probabilities of a customer being active in each time period. The probabilities thus obtained can then be used to calculate CLV.

2) *Extended application of Pareto/NBD model*

Reinartz and Kumar present a structured framework for identifying the customer lifetime profitability pattern in a non-contractual context [7]. A non-contractual setting

implies a setting where the type of product, quantity of product purchased, and timing of purchase depend completely on the customer and are uncertain from the point of view of the seller. The authors use their framework to test four commonly stated propositions regarding the profitability of long lifetime customers and conclude the following:

- a) *The relationship between customer lifetime and profitability is positive, but weak.*
- b) *No support for the argument that profits from long life customers increase over time.*
- c) *The notion that customers with long tenure are associated with lower promotional costs is rejected.*
- d) *Long lifetime customers do not pay higher prices.*

C. Normative models of CLV

Normative models provide us an opportunity to explore this issue without the “noise” encountered by empirical studies. Such models provide valuable insight for policy-making.

Blattberg and Thomas proposed a fundamental equation of customer equity [8], as follows:

$$CE(t) = \sum_{i=0}^I N_{i,t} \alpha_{i,t} (S_{i,t} - c_{i,t}) - N_{i,t} B_{i,\alpha,t} + \sum_{k=1}^{\infty} N_{i,t} \alpha_{i,t} \rho_{i,y+k} (S_{i,t+k} - c_{i,t+k} - B_{i,r,t+k} - B_{i,AO,t+k}) \left(\frac{1}{1+d} \right)^k$$

$$CE = \sum_{k=0}^t CE(t-k)$$

where $CE(t)$ = the customer equity value for customers acquired at time t ; $N_{i,t}$ = the number of potential customers at time t for segment i ; $\alpha_{i,t}$ = the acquisition probability at time t for segment i ; $\rho_{i,t}$ = the retention probability at time t for a customer for segment i ; $B_{i,\alpha,t}$ = the marketing cost per prospect for acquiring customers at time t for segment i ; $B_{i,r,t}$ = the marketing costs in time period t for retained customers for segment i ; $B_{i,AO,t}$ = the marketing costs in time period t for add-on selling for segment i ; d = the discount rate; $S_{i,t}$ = Sales of the product/services offered by the firm at time t for segment i ; $c_{i,t}$ = Cost of goods at time t for segment i ; I = the number of segments; i = the segment designation; t_0 = the initial time period; k = the time since acquisition.

The authors propose the use of this model to analyze the effects of elements of the marketing mix on the long-term value of the firm's customer base. They propose using customer equity principles to analyze and manage the marketing function.

III. CONCLUSION

Customer relationship management allows firms to distinguish between customers that are profitable, nearly profitable, unprofitable, and have the potential to be profitable. This kind of knowledge can help firms make focused and more effective decisions. It is common knowledge that loyal customers stay longer with a firm and are more profitable because they are cheaper to serve, they

pay higher prices, and they refer new customers, and so on. However, most of these propositions have not been tested rigorously through research. A few research papers that do address these propositions find contradictory evidence, or conditional support. It will be interesting to see how future research shapes our perception of customer loyalty and subsequently of CLV.

Using the appropriate models helps management to assess more accurately CLV, thus creating a better foundation for strategy formulation in relationships. Managers can substantially improve their return on marketing investments by using a dynamic, customer-level measure of CLV for scoring rather than using the other metrics suggested in the literature.

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